Received: 7 February 2022 | Accepted: 13 September 2022

DOI: 10.1111/jdv.18593

JEADV JOURNAL OF THE EUROPEAN ACADEMY OF DERMATOLOGY & VENEREOLOGY

POSITION STATEMENT

Quality of life measurement in vitiligo. Position statement of the European Academy of Dermatology and Venereology Task Force on Quality of Life and Patient Oriented Outcomes with external experts

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Abstract

Members of the European Academy of Dermatology and Venereology (EADV) Task Force on Quality of Life (QoL) and Patient Oriented Outcomes reviewed the instruments available for health-related (HR) QoL assessment in vitiligo and together with external vitiligo experts (including representatives of the EADV Vitiligo Task Force) have made practical recommendations concerning the assessment of QoL in vitiligo patients. The Dermatology Life Quality Index (DLQI) was the most frequently used HRQoL instrument, making comparison of results between different countries possible. Several vitiligo-specific instruments were identified. The vitiligo Impact Scale (VIS) is an extensively validated vitiligo-specific HRQoL instrument with proposed minimal important change and clinical interpretation for VIS-22 scores. VIS-22 was developed for use in India, where there are some specific cultural beliefs concerning vitiligo. The EADV Task Force on QoL and Patient Oriented Outcomes recommends use of the DLQI and the Children's Dermatology Life Quality Index (CDLQI) as dermatology-specific instruments in vitiligo. There is a strong need for a valid (including cross-cultural validation) vitiligo-specific instrument that can be either a new instrument or the improvement of existing instruments. This validation must include the proof of responsiveness.

INTRODUCTION

Vitiligo is the most common depigmenting skin disorder, with an estimated prevalence of 0.5%–2% in both adults and children worldwide. Males and females are equally affected, although women and girls seek consultation more frequently. The treatment of vitiligo is still one of the most difficult dermatological challenges.[1] Vitiligo may be the source of severe psychological distress, diminished quality of life (QoL) and increased risk of psychiatric morbidity.[2] Patients with darker skin types report higher health-related QoL (HRQoL) impairment than those with fairer skin types.[3] Cultural and religious influences result in varying attitudes to vitiligo in different countries.[4] There is often a need for a psychoeducational intervention with focus on acceptance and managing social impact.[5] To assess the impact of vitiligo on patients' lives, several vitiligo-specific instruments have been developed in different countries.[6]

The European Academy of Dermatology and Venereology (EADV) Task Force on QoL and Patient Oriented Outcomes has previously presented recommendations on principles of HRQoL instrument selection and use in different skin diseases.[7–20] Members of this Task Force decided to review the instruments available for HRQoL assessment in vitiligo and to make practical recommendations concerning the assessment of QoL in vitiligo patients.

METHODS

Members of the EADV Task Force on QoL and Patient Oriented Outcomes and external vitiligo experts (including representatives of the EADV Vitiligo Task Force) were invited to participate. A literature search was performed using the PubMed database, which was searched from 1990 to October 2021 using the key word combination: 'quality of life, vitiligo'. This review was planned to support the Position Statement of the EADV Task Force on Quality of Life and Patient Oriented Outcomes, and not planned as a systematic review. All those who volunteered were allocated a section of the identified articles to review.

Inclusion criteria

All identified publications written in English or having English abstracts were considered.

Exclusion criteria

- Review articles, guidelines, protocols
- Studies without HRQoL assessment
- Measurement of HRQoL in conditions other than vitiligo
- Studies where HRQoL was studied in vitiligo and other diseases but results on vitiligo were not presented and/or discussed separately

All publications were independently assessed by two co-authors. The results of literature search assessments were compared and discrepancies discussed and resolved. The remaining publications were analysed in detail, and the QoL instruments used in vitiligo were listed. The EADV Task Force on QoL and Patient Oriented Outcomes recommends using the word 'quimp'[21] (quality of life impairment) in routine clinical work and research,[22] and the word has been used in this article.

RESULTS

From the 378 articles identified in the literature search, 234 were excluded based on the exclusion criteria or duplication, leaving 144 publications for the final analysis (Table S1).

QoL instruments used

Generic, dermatology-specific and vitiligo-specific instruments were used in the identified publications. The most frequently used instrument (91 times) was the dermatology-specific Dermatology Life Quality Index (DLQI).[23] Ten instruments were used more than once (Figure 1). Several modifications of well-validated HRQoL instruments were used, with varying levels of revalidation. Detailed descriptions of identified vitiligo-specific instruments are presented in Table 1.

QoL impairment in patients with different locations of vitiligo lesions

Several studies reported worse HRQoL in patients with vitiligo located on visible sites (e.g. face and hands), on genitalia or with disseminated lesions.[33–41] However in contrast, in two studies vitiligo on visible locations did not interfere with HRQoL and there was no difference found between vitiligo patients with or without genital involvement.[42,43]

QoL in patients with vitiligo from different ethnicities and of different skin tones

Patients with darker skin types reported higher DLQI scores than those with fairer skin types.[3] In a study from Malaysia, Malay patients with vitiligo had significantly higher DLQI scores than Indians.[39] Higher quimp in patients with darker skin types than in those with fair skin was also reported.[42] Mean DLQI scores of adult vitiligo patients from studies performed in 21 different countries are presented in Figure 2.[44–83] Most studies demonstrated a moderate or small effect on patient's lives score according to the DLQI banding descriptors. Children's Dermatology Life Quality Index (CDLQI) mean scores of children with vitiligo from studies performed in several different countries are presented in Figure 3.[84–91] Most studies demonstrated a small effect on children's lives according to the CDLQI score banding descriptors.

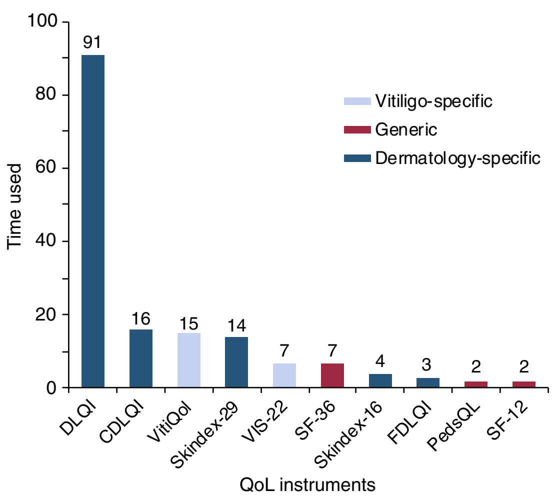

FIGURE 1 How many times HRQoL instruments were used in vitiligo

QoL in vitiligo patients of different ages

Only 4.1% of 15- to17-year-old teenagers reported no impact from their vitiligo, compared to 45.6% of children aged 0–6 years and 50.0% of children aged 7–14 years.[84] There was a significant correlation between the child's age and HRQoL impairment.[85] In another study, HRQoL was significantly worse in adolescents.[41] Young patients showed higher impairment of QoL and also higher levels of depression.[44] Dermatology-specific QoL was more impaired in those who were younger than 30 years old than in older subjects.[92] The mean DLQI score was higher in the 30–59 years old group than in the over 60 years old group.[39]

Gender differences of QoL in vitiligo

In several studies, there were no statistically significant differences in QoL scores between males and females.[39,49,57,58] However, in many other studies, HRQoL was significantly worse in female patients.[41,54,85,92–96] In one study, females experienced a greater impact on feelings and men experienced a greater impact on relationships.[38] In another study, women reported having more emotional problems with their vitiligo than men.[42]

QoL impairment in married and single patients with vitiligo

Several studies reported higher quimp in unmarried and divorced patients with vitiligo than in married patients.[58,59,95,97] One study showed no significant difference between DLQI scores of married and unmarried vitiligo patients.[39]

QoL in vitiligo and other diseases

In many studies, QoL in children and adults with vitiligo was shown to be substantially more impaired than in atopic dermatitis and psoriasis.[60,61,62,89,98–100] Higher total Skindex-16 scores were observed for patients with urticaria and acne than those with vitiligo.[101] One study reported greater quimp in vitiligo patients compared to those with melasma,[63] but no significant difference was found in other

TABLE 1 Information on vitiligo-specific HRQoL instruments

Instrument	Number of items	Domains	Scoring	Recall period	Validation
Vitiligo Quality of Life (VitiQoL)[24]	15 + one item for personal assessment of the severity of vitiligo, which corresponds to the 16th question of the VitiQoL questionnaire	3 domains: participation limitation, stigma and behaviour	15 item scores from 0 (never) to 6 (all the time). It yields a total score from 0 to 90. Single item on vitiligo severity scores from 0 (no skin involvement) to 6 (worst case). Higher scores indicate a more severe HRQoL impairment	Last month	Internal consistency, convergent validity, test–retest reliability (Brazilian version)[25]
Vitiligo Impact Scale (VIS) – 22[26]	22	10 domains: self-confidence, anxiety, depression, marriage, family worries, social interactions, school/college-related, occupation-related, treatment-related and attitude	'0' for 'not at all', '1' for 'a little', '2' for 'a lot' and '3' for 'very much'. The score ranged from 0 to 66 with higher scores indicating a higher impairment in the QoL	Not stated	Internal consistency, convergent validity, test–retest reliability. The minimal important change[27] and clinical interpretation for the VIS-22 scores were proposed by the authors of the instrument[28]
Vitiligo Life Quality Index (VLQI)[29]	25	6 domains: emotion, personal relationship, anxiety, work/school life, leisure and symptom	The answer to each question was scored as 'never = 1', 'sometimes = 2', 'often = 3' and 'all the time = 4', so the total score ranged between 25 and 100. Higher scores represented more severely impaired of QoL	Last week	Internal consistency, convergent validity, test–retest reliability
Vitiligo Impact Patient scale (VIPs)[30]	VIPs-fair skin (FS) – 22 items, VIPs- dark skin (DS) – 26 items. In total 29 items constituted the global VIPs questionnaire, of which 19 were found in both dark and fair skin phototypes, 3 were specific to fair skin, and 7 to dark skin	3 domains: psychological effects on daily life, relationships and sexuality, and economic constraints, care and management of disease	Responses to VIPs for each item were rated using a 6-point Likert scale: 'never' (rated 0), 'rarely' (1), 'sometimes' (2), 'often' (3), 'very often' (4) and 'constantly' (5). The VIPs can be reported as a total score (range, 0–110 for VIPs-FS and 0–130 for VIPs-DS). In these scores, 0 represents no effect and 110 maximal burden for fair-skinned patients and 130 maximal burden for dark-skinned patients. To allow a comparison between fair and dark skin phototype, authors proposed to report scoring as a score out of 100	Last 2 weeks	Internal consistency, convergent validity, test–retest reliability
Vitiligo Impact Patient scale -Short Form (VIPs-SF)[31]	VIPs-12-FS - 12 items and VIPs-12-DS - 12 items. In total 19 items constituted the global VIPs questionnaire, of which 5 were found in both dark and fair skin phototypes, 7 were specific to fair skin, and 7 to dark skin	3 domains: psychological effects on daily life, relationships and sexuality, and economic constraints, care and management of disease	Responses to VIPs for each item were rated using a 6-point Likert scale: 'never' (rated 0), 'rarely' (1), 'sometimes' (2), 'often' (3), 'very often' (4), and 'constantly' (5). The total score for the VIPs short-forms (VIPs-12-DS and VIPs12-FS) ranged from 0–60 for the 12 items	Last 2 weeks	No data
Family Vitiligo Impact Scale[32]	16	12 broad domains: disease understanding, treatment, social impact, impact on affect, financial burden, functionality, physical, functionality, occupational, functionality, leisure, familial conflicts, cognition, affected family member's functioning and interpersonal relationship	'0' for 'not at all', '1' for 'a little', '2' for 'a lot' and '3' for 'very much'. The score ranged from 0 to 48 with higher scores indicating a higher impairment in the QoL	Not stated	Internal consistency, convergent validity, test–retest reliability

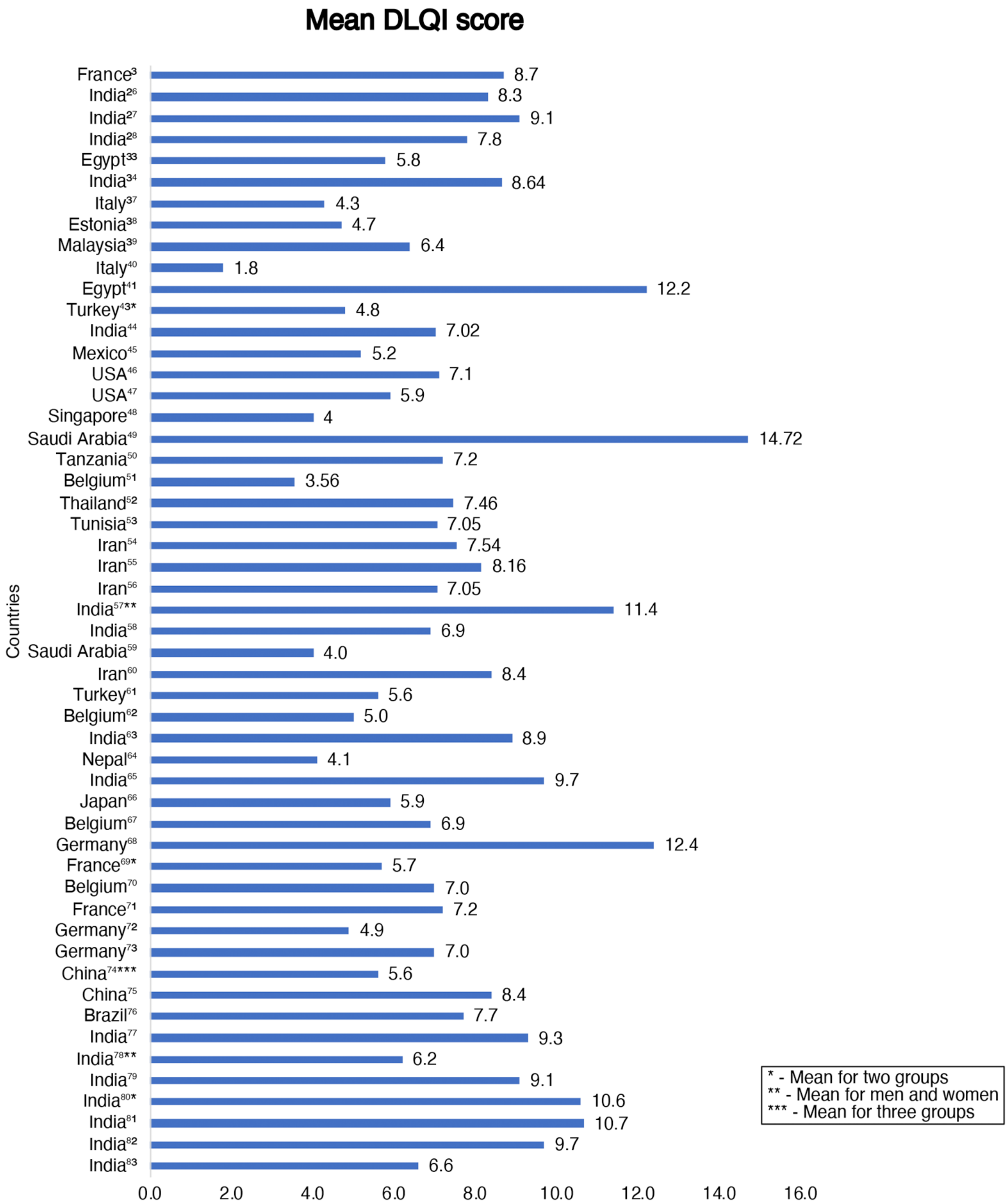

FIGURE 2 Mean DLQI scores in different studies

studies.[64,65] There was no significant difference between HRQoL of children with vitiligo and alopecia areata.[102]

QoL in people with vitiligo and comorbidities

Patients with diabetes reported a greater QoL impact and more depression and anxiety if they had vitiligo, compared to patients without skin problems.[103] Vitiligo-like lesions in patients with advanced breast cancer treated with cyclin-dependent inhibitors had a negative impact on their HRQoL.[104] Multiple regression analyses showed that vitiligo patients who had thyroid disease were more likely to report poor HRQoL than those without thyroid disease.[48] In the study by Sampogna et al.,[105] the association of QoL impairment with psychological problems was very strong.

Family QoL of patients with vitiligo

QoL of family members of patients with vitiligo was more impaired than the family QoL of controls.[106] QoL impairment of family members of children with vitiligo, measured using the Family Dermatology Life Quality Index (FDLQI), increased with the age of the children.[107] Use of camouflage improved the QoL not only of children with vitiligo but also of their parents.[86] In a study from Saudi Arabia family, QoL scores significantly correlated with patients' HRQoL scores, higher educational levels of family members and negatively

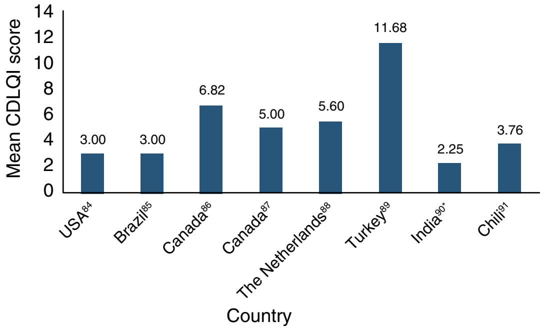

FIGURE 3 Mean CDLQI scores in different studies

correlated with duration of vitiligo.[108] No significant difference in parental QoL was found between patients with vitiligo and with alopecia areata.[102]

HRQoL in vitiligo clinical trials

There has been measurement of HRQoL in people with vitiligo using the following treatments: cosmetic camouflage,[41,66,67,87] narrowband UVB,[34,88,109–115] PUVA, PUVA sol,[116] excimer laser,[117,118] topical corticosteroids,[110,112,117] calcipotriol,[112,115] tacrolimus,[68,69] cyclosporine,[119] levamisole,[90] group climatotherapy[120] and noncultured epidermal cellular grafting.[70] Improvement of HRQoL was reported in all these studies. However, only cosmetic camouflage users showed significantly better HRQoL improvement than non-users,[41,66] and patients treated with PUVA reported better quimp than those treated by PUVA sol[116] in comparative studies.

DISCUSSION

In the vitiligo studies reviewed, the DLQI was the most frequently used HRQoL instrument and this has made it possible to compare results between different countries. Data in Figure 2 demonstrate that in non-European countries the reports are of patient groups with high DLQI scores, with some minor exceptions. Most studies showed worse HRQoL in vitiligo patients than in controls.[121]

Other dermatology-specific HRQoL instruments (Skindex-29, Skindex-16) were used much less often than the DLQI and therefore provide currently limited potential for comparison among different vitiligo studies. Score banding system and the minimal clinically important difference for the DLQI were established and validated.[122–123] The generic HRQoL instrument SF-36 may be used for comparison of vitiligo with non-dermatologic diseases but the number of such studies is low. Some researchers consider that neither generic nor dermatology-specific instruments reflect all aspects of QoL impairment in vitiligo because aspects more common or severely affected in vitiligo may not be a part of these instruments. Conversely, some items (such as physical symptoms) in these generic or general dermatology instruments are not applicable to vitiligo.[26] This may lead to lower total scores in vitiligo patients when compared to other skin diseases which might underestimate the disease burden of vitiligo. Comments on vitiligo-specific instruments are given below.

Items that are irrelevant for most of patients with vitiligo for example on itching, pain and bleeding are absent in the Vitiligo Quality of Life (VitiQoL).[24] This instrument contains the only item on sun protection which may be considered with some reservations as disease-specific. The rest of the items may be also relevant to skin diseases other than vitiligo. In contrast with other questions, in two items skin disease is not mentioned: 'While talking to someone, did you worry about what people might be thinking about you' and 'Were you afraid that people would criticize you'. As skin disease is not mentioned in these, it is possible that this might lead some patients to the misunderstanding that these two questions are not limited to the effect of vitiligo, but are more general. The VitiQoL questionnaire has a 1-month recall period. This may help to gather more long-term impact information than the use of instruments with a 1 or 2 weeks recall period, but the 4-week period makes it impossible to detect changes over shorter time intervals when assessing treatment effects. However, vitiligo is generally a slowly changing condition, so this may not be a disadvantage. The VitiQoL was used in 12 studies from different countries including four clinical trials. However, this score failed to show responsiveness in large randomized clinical tr ials.[36,45,85,93,109–111,119,124–127]

Vitiligo Impact Scale (VIS)–22[26] is a validated vitiligo-specific HRQoL instrument with a known minimal important score change[27] and clinical descriptive interpretation of VIS-22 scores.[28] However, this instrument was developed in India for Indian people where there are specific and strong cultural issues concerning vitiligo.[4] There is no clear recall period for VIS-22. Some items probably cover the whole period of having vitiligo and so the scores of such items should not change even after successful therapy. In several VIS-22 items, vitiligo is not mentioned and (as in case of the VitiQoL) this may lead to a misunderstanding that the questions are not limited of vitiligo. VIS-22 was used in four different studies including studies from China and Nepal.[44,65,92,125]

The Vitiligo Life Quality Index (VLQI)[29] has a 1-week recall period and so may reflect HRQoL changes after short periods of time, such as following treatment. In contrast to other vitiligo-specific instruments, the VLQI contains an item on pain, irritation or itching due to vitiligo. Unusually, in the scoring system the answer 'never' is scored as '1' but not '0'. Thus, absence of quimp in this 25 item questionnaire will still have a score of 25 rather than the more logical zero. We would advise the authors of the VLQI to consider changing this.

The Vitiligo Impact Patient scale (VIPs)[30] has different versions for patients with 'fair skin' (FS) (Fitzpatrick skin phototype I to III) and 'dark skin' (DS) (Fitzpatrick skin phototype IV to VI). The global VIPs questionnaire has 29 items, of which 19 were found in both the 'dark' and 'fair' skin versions. In addition, there are three questions specific to the FS version and seven to the DS version.

The authors of the VIPs considered that it would be difficult to use it efficiently in daily clinical care and therefore developed the Vitiligo Impact Patient scale-Short Form (VIPs-SF).[31] There are two versions of this 12 items scale, the FS (VIPs-12-FS) and the DS (VIPs-12-DS) versions. Of the 12 questions in the VIPs-SF, five are the same in both versions, seven are specific to VIPs-12-FS and seven to FS, and 7 to VIPs-12-DS. However, identical items appear in different domains in VIPs-12-DS and VIPs-12-FS. The items 'I tend to withdraw into myself because of my vitiligo' and 'I dread first meetings because of my vitiligo' are placed in 'Psychological effects on daily life' in the VIPs-12-DS and in 'Relationships and Sexuality' in the VIPs-12-FS. In clinical practice or research, the use of two different versions of the same instrument for patients with different phototypes is likely to be unwieldy and impractical.

The Family Vitiligo Impact Scale[32] was developed in India and would need to be revalidated prior to use in other countries, especially because of the specific cultural issues mentioned above. This questionnaire measures the secondary impact on the QoL of family members caused by having someone in the family with vitiligo.

Most studies reviewed recorded worse quimp in vitiligo patients with lesions on visible sites, in young, unmarried patients, in patients with fairer skin types and in females. To confirm this and to reduce the influence of confounding factors, it would be necessary to undertake large population studies or at least studies where participants are well matched across a range of factors, as in the studies of gender differences.[128,129]

HRQoL assessment in children is even more challenging than in adult patients.[130] The CDLQI has been used in eight studies to study HRQoL impairment in children with vitiligo.[41,84–88,131] The CDLQI[132] is a dermatology-specific instruments for self-assessment of the HRQoL in children aged 4 years or more. Scores banding system of the CDLQI was validated.[133] In younger children, the dermatology-specific proxy instrument the InToDermQoL may be used.[134–136]

The QoL impact on family members of a person with vitiligo was captured well by the FDLQI. However, the Family Vitiligo Impact Scale, developed in India, may better reflect family QoL changes in this regional context. However, in atopic dermatitis, it was shown that the FDLQI scores correlated well with a disease-specific family QoL instrument.[137]

This review has some limitations. There are implied comparisons of scores between different studies, but the inclusion/exclusion criteria and other selection biases may render such comparisons less accurate. The limiting of the selection of articles to the English language may have resulted in the omission of some relevant research articles.

There is current opinion, exemplified by the development of core outcome sets[138] that researchers, and clinicians should reach uniform decisions about which HRQoL measures to choose. This leads to benefits, especially in the ability to compare and understand scores across different centres. However, there may be disadvantages where HRQoL assessment data may be similar but not identical even in neighbouring countries[139,140] and where there are major cultural and geographic differences, especially in relation to vitiligo. The ability to interpret the scores of instruments and the defining of the minimal clinically important score differences for each instrument are important to allow HRQoL questionnaires to be of practical use, especially clinically.

In conclusion, the EADV Task Force on QoL and Patient Oriented Outcomes recommends use of the DLQI and the CDLQI as dermatology-specific instruments in patients with vitiligo. However, there is a strong need for a valid (including cross-cultural validation) vitiligo-specific instrument that can be either a new instrument or the improvement of existing instruments. Although we acknowledge that despite important efforts that have been conducted to develop vitiligo-specific HRQOL instruments, none of the existing instruments have been fully validated, especially for responsiveness. However, it seems important to say that although the existing instruments might be improved, it is worth to use them in the settings of clinical trials and in daily clinic rather than not to measure HRQoL or some of its aspects.

ACKNOWLEDGEMENTS

None.

CONFLICT OF INTEREST

AYF is joint copyright owner of the DLQI, CDLQI, IDQoL, DFI and FDLQI and other quality of life measures: Cardiff University receives royalties from the use of these measures: AYF receives a share under standard university policy. AYF has received honoraria for advisory boards: Novartis, USB Pharma and speaker honoraria from Lilly and Medscape. JCS served as a consultant for Abbvie, Leo Pharma, Novartis, Sandoz, Sanofi-Genzyme, Trevi, Viofor, Jannsen-Cilag and Eli-Lilly. CS reports non-financial support from Leo Pharma and Abbvie, Royalties from Springer AG, grants from Abbvie, Berlin Chemie, LeoPharma, CeraVe International, outside the submitted work. AB reports grant from Janssen, royalties or licences from Wiley, consulting fees from Abbvie, Almirall, Galderma, Eli Lilly, Janssen, Novartis, Leo Pharma, Sanofi and UCB, payment or honoraria from Abbvie, Almirall, Galderma, Eli Lilly, Janssen, Novartis, Leo Pharma, Sanofi and UCB. JS has received grants and/or honoraria from AbbVie, Calypso Biotech, Bristol Myers Squibb, Incyte Corporation, LEO Pharma, Eli Lilly, Novartis, Pfizer, Pierre-Fabre, Sanofi, Sun Pharmaceuticals and Viela Bio; and has patents on MMP9 inhibitors and uses thereof in the prevention or treatment of a depigmenting disorder, and three-dimensional model of depigmenting disorder. Other authors reported no conflicts of interests.

DATA AVAILABILITY STATEMENT

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Chernyshov PV, Tomas-Aragones L, Manolache L, Pustisek N, Salavastru CM, Marron SE, et al. Quality of life measurement in vitiligo. Position statement of the European Academy of Dermatology and Venereology Task Force on Quality of Life and Patient Oriented Outcomes with external experts. J Eur Acad Dermatol Venereol. 2023;37:21–31. https://doi.org/10.1111/jdv.18593